Fundamental theorems of calculus, integration-by-parts, and weak derivatives

calculations for weeks 10 & 11

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UAF Math 617 Functional Analysis

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Outline

1. the fundamental theorem of calculus in $\mathbb{R}^1$
2. divergence theorem on $\mathbb{R}^d$
3. integration by parts (in $\mathbb{R}^1$ and in $\mathbb{R}^d$)
4. weak derivatives
5. lecture content in weeks 10 & 11

indefinite integral of an L^1 function

- for $a < b$ real numbers, recall that

$$L^1([a,b]) = \left\{ f : [a,b] \to \mathbb{R} \,\middle|\, f \text{ is measurable and } \int_a^b |f(x)|\,dx < \infty \right\}$$

    - remember this is actually a *lie*; the elements of L^1 are *equivalence classes* up to almost everywhere
- what happens when you integrate $f \in L^1$ to get a new function?

$$F(x) = \int_a^x f(t)\,dt$$

    - this is an *indefinite integral*
- does the fundamental theorem of calculus (FTC) apply?

$$\frac{d}{dx}\left(\int_a^x f(t)\,dt\right) \stackrel{?}{=} f(x)$$

    - is $F'(x) = f(x)$?

Theorem (easy direction)

If $f \in L^1([a,b])$ then $F(x) = \int_a^x f(t)\,dt$ is continuous, and it is differentiable almost everywhere, and $F' = f$ almost everywhere.

picture of proof:

fundamental theorem of calculus (FTC)

Theorem (easy direction)

If $f \in L^1([a,b])$ then $F(x) = \int_a^x f(t)\,dt$ is continuous, and it is differentiable almost everywhere, and $F' = f$ almost everywhere.

Proof when we also assume f is continuous.

If $f \in C([a,b])$ then bounded: $|f| \le M$. Let $\epsilon > 0$. If $|x-y| < \delta = \frac{\epsilon}{M}$, and $x<y$, then

$$|F(x) - F(y)| = \left|\int_x^y f(t)\,dt\right| \le \int_x^y |f(t)|\,dt \le M|x-y| < M\delta = \epsilon,$$

so F is continuous. Since f is continuous at x then there is $\gamma = \gamma(x) > 0$ so that $|f(t) - f(x)| < \epsilon$ if $|t-x| < \gamma$. Choose any $0 < h < \gamma$. Then

$$\begin{aligned}\left|\frac{F(x+h)-F(x)}{h} - f(x)\right| &= \left|\frac{1}{h}\int_x^{x+h} f(t)\,dt - \frac{1}{h}\int_x^{x+h} f(x)\,dt\right| \\ &\le \frac{1}{h}\int_x^{x+h} |f(t)-f(x)|\,dt < \frac{1}{h}\int_x^{x+h} \epsilon\,dt = \epsilon\end{aligned}$$

(A similar argument handles $-\gamma < h < 0$.) This shows $f(x) = \lim_{h\to 0} \frac{F(x+h)-F(x)}{h}$; in particular the two-sided limit exists at every x. $\square$

how the integral is well-behaved when $f \in L^1$

Lemma ("absolute continuity of the Lebesgue integral")

Suppose $f \in L^1([a,b])$ and $\epsilon > 0$. There exists $\delta > 0$ so that if $m(E) < \delta$ then

$$\int_E |f|\, dm < \epsilon.$$

Proof.

Let $g_k = \min\{|f|, k\}$. This defines a sequence $0 \le g_k \le |f|$ so that $g_k \nearrow |f|$, but where $0 \le g_k \le k$ so g_k is bounded. By the Monotone Convergence Theorem,

$$\int_{[a,b]} |f|\, dm = \lim_{k\to\infty} \int_{[a,b]} g_k\, dm$$

Given $\epsilon > 0$, choose K large enough that $\int_{[a,b]} (|f| - g_K)\, dm < \epsilon/2$. Then for any measurable $E \subset [a,b]$:

$$\int_E |f|\, dm = \int_E g_K\, dm + \int_E (|f| - g_K)\, dm \le \int_E K\, dm + \int_{[a,b]} (|f| - g_K)\, dm \le K \cdot m(E) + \frac{\epsilon}{2}$$

Now set $\delta = \epsilon/(2K)$. If $m(E) < \delta$ then $\int_E |f|\, dm < \epsilon$. □

picture of the absolute continuity of the Lebesgue integral

picture:

classical definition of absolute continuity

Definition

a continuous function $f : [a, b] \to \mathbb{R}$ is *absolutely continuous* if for all $\epsilon > 0$ there is $\delta > 0$ so that if we have a finite collection of disjoint intervals $[\alpha_i, \beta_i]$, with total length less than δ:

$$\sum_{i=1}^{n} (\beta_i - \alpha_i) < \delta,$$

then the variation of f on those intervals is less than ϵ:

$$\sum_{i=1}^{n} |f(\alpha_i) - f(\beta_i)| < \epsilon$$

picture of absolute continuity

picture:

flavors of continuity for $f : [a, b] \to \mathbb{R}$

C^1

$\Downarrow$

Lipschitz continuous

$\Downarrow$

absolutely continuous

$\Downarrow$

uniformly continuous

$\Downarrow$

C^0 (continuous at each x)

Definition

a continuous function $f : [a, b] \to \mathbb{R}$ is *Lipschitz continuous* if there is $L \geq 0$ so that

$$|f(x) - f(y)| \leq L|x - y|$$

for all $x, y \in [a, b]$

Theorem (FTC 1)

If $f \in L^1([a,b])$ then $F(x) = \int_a^x f(t)\,dt$ is absolutely continuous, and it is differentiable almost everywhere, and $F' = f$ almost everywhere.

sketch of the proof.

Approximate f in L^1 by a continuous function g. Show that the difference $f - g$, which can be large, can't be large on a big set. (Vitali covering lemma.) Conclude. $\square$

Theorem (FTC 2)

If $F(x)$ is absolutely continuous then $f = F'$ exists a.e. and $f \in L^1$ and $F(x) - F(a) = \int_a^x f(t)\,dt$ for all x.

- take-home message: if $f \in L^1$ then its indefinite integral $F(x) = \int_a^x f(t)\,dt$ is a nice "very" continuous function which is differentiable a.e.
- *and* you can recover f from F by differentiating, at least a.e.

not the FTC, part 1: continuity is necessary ...

question: if $F : [a, b] \to \mathbb{R}$ is differentiable almost everywhere ($f = F'$ a.e.), and if $f \in L^1$, then is it true that

$$F(x) - F(a) = \int_a^x f(t)\,dt \quad \text{for all } x \in [a, b]?$$

answer: no

picture:

- my pair of formulas with $F'(x) = G'(x)$ a.e.:

$$F(x) = \begin{cases} x, & 0 < x < 1 \\ 2, & 1 < x < 2 \end{cases} \qquad G(x) = \begin{cases} x, & 0 < x < 1 \\ 1, & 1 < x < 2 \end{cases}$$

$$F'(x) = G'(x) = \begin{cases} 1, & 0 < x < 1 \\ 0, & 1 < x < 2 \end{cases}$$

question: if $F : [a, b] \to \mathbb{R}$ is continuous, and if $f = F'$ exists almost everywhere, and if $f \in L^1$, then is it true that

$$F(x) - F(a) = \int_a^x f(t)\,dt \quad \text{for all } x \in [a, b]?$$

answer: also no!

picture:

final conclusion: Lebesgue's FTC in 1D

Fundamental Theorem of Calculus

1. If $f \in L^1([a,b])$ then $F(x) = \int_a^x f(t)\,dt$ is in $AC([a,b])$,[a] and it is differentiable a.e., and
$$\frac{d}{dx}\left(\int_a^x f(t)\,dt\right) = f(x)$$
for almost every x.
2. If $F \in AC([a,b])$ then $f = F'$ exists a.e., and $f \in L^1([a,b])$, and
$$F(x) - F(a) = \int_a^x f(t)\,dt$$
for all x.

[a] We define $AC([a,b])$ = (absolutely continuous functions on $[a,b]$).

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divergence theorem in the plane $\mathbb{R}^2$

Theorem (divergence theorem . . . as a Green's theorem)

Suppose $\Omega \subset \mathbb{R}^2$ *is bounded and open. Suppose that the curve* $C = \partial\Omega$ *is continuously-differentiable, and orient it counter-clockwise. If* $u(x, y)$ *and* $v(x, y)$ *are continuously-differentiable functions on* $\overline{\Omega}$ *then*

$$\iint_\Omega u_x + v_y \, dx\, dy = \oint_C u\, dy - v\, dx.$$

- the right side is a *curve (line) integral*; we must parameterize C as $(x(t), y(t))$:

$$\oint_C u\, dy - v\, dx = \int_0^T u(x(t), y(t))y'(t) - v(x(t), y(t))x'(t)\, dt$$

- pictures (with disconnected and not-simply-connected possibilities):

Theorem (divergence theorem)

Suppose $\Omega \subset \mathbb{R}^2$ is bounded and open, and $\partial\Omega$ is continuously-differentiable. Let $\mathbf{n}$ be the outward normal along $\partial\Omega$. If $\mathbf{V}$ is a continuously-differentiable vector field on $\overline{\Omega}$ then

$$\int_\Omega \nabla \cdot \mathbf{V}\, dm = \int_{\partial\Omega} \mathbf{V} \cdot \mathbf{n}\, ds$$

- $\mathbf{V} = \mathbf{V}(x, y)$ has components $\mathbf{V} = \langle u, v \rangle$
- $\nabla \cdot \mathbf{V} = u_x + v_y$ is the *divergence* of the vector field
- I will just use "$\int$" instead of $\iint$, $\oint$, etc.
- dm is Lebesgue measure in $\mathbb{R}^2$
- ds is (induced submanifold) Lebesgue measure along the 1D curve $\partial\Omega$
  - nontrivially defined! use parameterized boundary, Jacobian determinant, ...

Example

Let $\Omega = \{(x,y) \,:\, x^2 + y^2 < 1\}$ be the unit disk. Then $\partial\Omega$ is the unit circle. Let $\mathbf{V} = \langle x, y+1 \rangle$. Is it true?

$$\int_\Omega \nabla \cdot \mathbf{V}\, dm \stackrel{?}{=} \int_{\partial\Omega} \mathbf{V} \cdot \mathbf{n}\, ds$$

solution steps:

- parameterize $\partial\Omega$: $(x(t), y(t)) = (\cos(t), \sin(t))$, $0 \le t \le 2\pi$
- $\mathbf{n} = \langle x, y \rangle = \langle \cos(t), \sin(t) \rangle$
- $ds = dt$ (arclength parameterized)
- $\mathbf{V} \cdot \mathbf{n} = \langle x, y+1 \rangle \cdot \langle x, y \rangle = x^2 + y^2 + y = 1 + \sin(t)$
- $\nabla \cdot \mathbf{V} = 1 + 1 = 2$
- thus (*no surprise!*) the two sides are equal:

$$\int_\Omega \nabla \cdot \mathbf{V}\, dm = \int_\Omega 2\, dm = 2\pi$$

$$\int_{\partial\Omega} \mathbf{V} \cdot \mathbf{n}\, ds = \int_0^{2\pi} 1 + \sin(t)\, dt = 2\pi$$

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divergence theorem in $\mathbb{R}^d$

- there is nothing intrinsically two-dimensional about the theorem!

Theorem (divergence theorem)

Suppose $\Omega \subset \mathbb{R}^d$ is bounded and open, and $\partial\Omega$ is continuously-differentiable or Lipschitz. Let $\mathbf{n}$ be the outward normal along $\partial\Omega$. If $\mathbf{V}$ is a continuously-differentiable vector field on $\overline{\Omega}$ then

$$\int_\Omega \nabla \cdot \mathbf{V}\, dm = \int_{\partial\Omega} \mathbf{V} \cdot \mathbf{n}\, ds$$

- dm is Lebesgue measure in $\mathbb{R}^d$
- ds is $(d-1)$-dim'l Lebesgue measure (submanifold measure)
- a point $x \in \mathbb{R}^d$ has *coordinates* x_i: $\quad x = (x_1, \dots, x_d)$
- a vector field $\mathbf{V} : \Omega \to \mathbb{R}^d$ has *components*, $\mathbf{V} = \langle v_1, \dots, v_d \rangle$, and the components are scalar functions, $v_k = v_k(x_1, \dots, x_d)$
- the *divergence* of the vector field is

$$\nabla \cdot \mathbf{V} = \frac{\partial v_1}{\partial x_1} + \dots + \frac{\partial v_d}{\partial x_d} = \sum_{k=1}^{d} \frac{\partial v_k}{\partial x_k}$$

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picture of divergence theorem in $\mathbb{R}^3$ (and hints of proof)

picture in $\mathbb{R}^3$:

most-famous usage in physics

- Gauss' law for electric fields might look like this in a physics text.

Gauss' law ("integral form")

Suppose $\mathbf{E} : \mathbb{R}^3 \to \mathbb{R}^3$ is the (continuously-differentiable) electric field. Suppose $\Omega \subset \mathbb{R}^3$ is a bounded domain with smooth boundary $\partial\Omega$. Then

$$\boxed{\oiint_{\partial\Omega} \mathbf{E} \cdot \mathbf{n}\, ds = \frac{Q}{\epsilon_0}}$$

where Q is the net enclosed charge within Ω, and $\epsilon_0 > 0$ is a constant.

- apply the divergence theorem on the left, and write Q as the integral of a charge density function $\rho : \mathbb{R}^3 \to \mathbb{R}$, $\rho \in L^1$:

$$\int_\Omega \nabla \cdot \mathbf{E}\, dm = \frac{1}{\epsilon_0} \int_\Omega \rho\, dm$$

- since this applies over any $\Omega \subset \mathbb{R}^3$, we get Gauss' law in "PDE form":

$$\boxed{\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}}$$

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fundamental theorems in any dimension

- the Fundamental Theorem of Calculus, in any dimension, says that the integral of a derivative can be computed by integrating over the boundary:

$$\int_\Omega d\omega = \int_{\partial\Omega} \omega \qquad \textit{(Stokes theorem)}$$

    - making this precise requires defining *chains* Ω and *differential forms* ω
    - which I'm not going to do
- a stack of Fundamental Theorems for $\Omega \subset \mathbb{R}^d$, in various dimensions:

$$\int_a^b F'\,dm = F(b) - F(a) \qquad \text{(interval domain, two-point boundary)}$$

$$\iint_\Omega \nabla\cdot\mathbf{V}\,dm = \oint_{\partial\Omega} \mathbf{V}\cdot\mathbf{n}\,ds \qquad \text{(planar domain, closed-curve boundary)}$$

$$\iiint_\Omega \nabla\cdot\mathbf{V}\,dm = \oiint_{\partial\Omega} \mathbf{V}\cdot\mathbf{n}\,ds \qquad \text{(solid domain, closed-surface boundary)}$$

$$\int_\Omega \nabla\cdot\mathbf{V}\,dm = \int_{\partial\Omega} \mathbf{V}\cdot\mathbf{n}\,ds \qquad (\mathbb{R}^d \text{ domain, } (d-1)\text{-dim'l closed boundary})$$

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product rule and integration-by-parts in $\mathbb{R}^1$

- this is very familiar!
- suppose $f(x), g(x)$ are continuously-differentiable (or absolutely continuous)
- then:

$$\Big(f(x)g(x)\Big)' = f'(x)g(x) + f(x)g'(x)$$

- if we integrate both sides from a to b, and apply FTC on left:

$$\Big[f(x)g(x)\Big]_a^b = \int_a^b f'(x)g(x)\,dx + \int_a^b f(x)g'(x)\,dx$$

- rearranging, we get integration-by-parts:[1]

$$\boxed{\int_a^b f(x)g'(x)\,dx = \Big[f(x)g(x)\Big]_a^b - \int_a^b f'(x)g(x)\,dx}$$

- it suffices for f, g to be absolutely continuous (and thus $f', g' \in L^1$)

[1] also known as $\int u\,dv = u\,v - \int v\,du$, but then you put in limits of integration ...

product rules in $\mathbb{R}^d$

- notation:
  - $x \in \mathbb{R}^d$ has coordinates: $x = (x_1, \dots, x_d)$
  - $f, g : \mathbb{R}^d \to \mathbb{R}$ are scalar functions
  - $\mathbf{V}$ has components: $\mathbf{V} = \langle v_1, \dots, v_d \rangle$ where v_k are scalar functions
  - gradient $\nabla f = \left\langle \frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_d} \right\rangle$ and divergence $\nabla \cdot \mathbf{V} = \sum_{k=1}^d \frac{\partial v_k}{\partial x_k}$
- there are 2 priority product rules in $\mathbb{R}^d$ to know:[2]

product rule 1: partial derivatives (of scalar functions)

$$\frac{\partial}{\partial x_k}\Big(f\, g\Big) = \frac{\partial f}{\partial x_k} g + f \frac{\partial g}{\partial x_k}$$

product rule 2: divergence/gradient

$$\nabla \cdot (f\, \mathbf{V}) = \nabla f \cdot \mathbf{V} + f\, \nabla \cdot \mathbf{V}$$

[2]there are also product rules for the curl ("$\nabla \times \mathbf{V}$"), and/or for the exterior derivative and differential-forms wedge product ("d" and "$\omega \wedge \nu$"); know these for electricity and magnetism!

- the 2 priority integration-by-parts rules in $\mathbb{R}^d$ come from the two product rules ... via the same logic as in $\mathbb{R}^1$!
- the first rule is simplest and most useful *when the boundary integral is zero*, thus the assumption of compact support
- because of the assumption of compact support, the boundary of Ω is allowed to be weird

integration-by-parts 1: partial derivatives

if $\Omega \subset \mathbb{R}^d$ is open, and if $f, g \in C_c^1(\Omega)$ then

$$\int_\Omega \frac{\partial f}{\partial x_k} g \, dm = -\int_\Omega f \frac{\partial g}{\partial x_k} \, dm$$

- but what is "$C_c^1(\Omega)$"?

Definition

- If $\Omega \subset \mathbb{R}^d$ is open, we say that a function $f : \Omega \to \mathbb{R}$ *has compact support* if

$$\operatorname{supp}(f) = \overline{\{x \in \Omega \,:\, f(x) \ne 0\}}$$

  is a compact subset of Ω.
- $C_c(\Omega) = \{\mathbb{R}\text{-valued continuous functions which have compact support}\}$
- $C_c^n(\Omega) = \{f \in C_c(\Omega) \,:\, \text{partial derivatives up to order } n \text{ are in } C_c(\Omega)\}$
- $C_c^\infty(\Omega) = \{f \in C_c(\Omega) \,:\, \text{partial derivatives of all orders are in } C_c(\Omega)\}$

- note $C_c^0(\Omega) = C_c(\Omega)$
- $C_c^\infty(\Omega)$ functions are the nicest! you can differentiate and/or integrate over Ω as much as you want

- the second formula, easily derived from the divergence theorem, is sometimes called a *Green's identity* or similar
- it requires the boundary to have a normal direction, so the boundary cannot be too weird

integration-by-parts 2: divergence/gradient

if $\Omega \subset \mathbb{R}^d$ is open, bounded, and has continuously-differentiable or Lipschitz boundary, and if f and the components of $\mathbf{V}$ are in $C^1(\overline{\Omega})$, then

$$\int_\Omega f \, \nabla \cdot \mathbf{V} \, dm = \int_{\partial\Omega} f \, \mathbf{V} \cdot \mathbf{n} \, ds - \int_\Omega \nabla f \cdot \mathbf{V} \, dm$$

compact support and integration-by-parts

pictures:

left: for partial derivatives

right: for divergence/gradient

application of integration-by-parts: weak form of a PDE

- suppose we want to solve the *Poisson equation*, a partial differential equation (PDE), on a domain $\Omega \subset \mathbb{R}^d$, with given data $f \in L^1(\Omega)$:

$$-\nabla^2 u \overset{*}{=} f$$

  - we seek $u : \Omega \to \mathbb{R}$, which also satisfies some boundary conditions (below)
  - notation: $\nabla^2 u = \nabla \cdot (\nabla u)$ is the *Laplacian* of u; it is also a scalar function
- concept: $*$ is true if and only if arbitrary integrals of the sides are equal
- multiply both sides of $*$ by another function v, a *test function*, expanding the Laplacian, and integrate over Ω:

$$-\int_\Omega \nabla \cdot (\nabla u) v\, dm = \int_\Omega f v\, dm$$

- apply integration-by-parts 2 for divergence/gradient:

$$\int_\Omega \nabla u \cdot \nabla v\, dm - \int_{\partial\Omega} v \nabla u \cdot \mathbf{n}\, ds = \int_\Omega f v\, dm$$

application of integration-by-parts: weak form of a PDE

- from last slide, Poisson's equation $-\nabla^2 u = f$ has become

$$\int_\Omega \nabla u \cdot \nabla v \, dm - \int_{\partial\Omega} v \nabla u \cdot \mathbf{n} \, ds = \int_\Omega f v \, dm$$

- the boundary term currently involves the solution u
- however, it becomes data if either
  1. we set the normal derivative, $\nabla u \cdot \mathbf{n} = g_N$ (*Neumann condition*), or
  2. we set $u = g_D$ and $v = 0$ along the boundary (*Dirichlet condition*)
- if we choose homogeneous Dirichlet conditions ($v = 0$ and $u = 0$ along boundary) then, essentially, we have equivalent forms of Poisson's equation:

$$\boxed{-\nabla^2 u = f} \qquad \Longleftrightarrow \qquad \boxed{\int_\Omega \nabla u \cdot \nabla v \, dm = \int_\Omega f v \, dm}$$

strong form   weak form

- the weak form is what we will use for finite element approximations

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simplifying awkward partial derivative notation

- integration by parts will allow us to differentiate very general functions
- but first: multi-indices simplify notation
- let $\mathbb{Z}^+ = \{0, 1, 2, \dots\} = \{0\} \cup \mathbb{N}$
- suppose $\alpha = (\alpha_1, \dots, \alpha_d)$ is a multi-index:

$$\alpha \in \mathbb{Z}^+ \times \dots \times \mathbb{Z}^+ = \left(\mathbb{Z}^+\right)^d,$$

- $|\alpha| = \alpha_1 + \dots + \alpha_d$ is the *order* of α

Definition

$$D^\alpha = \frac{\partial^{|\alpha|}}{\partial {x_1}^{\alpha_1} \dots \partial {x_d}^{\alpha_d}}$$

- for example on $f : \mathbb{R}^3 \to \mathbb{R}$:

$$\alpha = (3, 0, 0) \qquad \to \qquad D^\alpha f = \frac{\partial^3 f}{\partial {x_1}^3} \quad \text{(3rd-order)}$$

$$\alpha = (0, 1, 1) \qquad \to \qquad D^\alpha f = \frac{\partial^2 f}{\partial x_2 \partial x_3} \quad \text{(2nd-order mixed)}$$

- suppose $\Omega \subset \mathbb{R}^d$ is open
- for any function $\varphi : \Omega \to \mathbb{R}$, recall that we defined the closed set

$$\operatorname{supp}(\varphi) = \overline{\{x \in \Omega \,:\, \varphi(x) \neq 0\}}$$

- recall that for $C_c^\infty(\Omega)$ functions we may differentiate as much as we want
- if $\varphi \in C_c^\infty(\Omega)$ then the support of φ is compact
- if $\varphi \in C_c^\infty(\Omega)$ then the support of *any* $D^\alpha\varphi$, for any α, is compact

main idea:

if $\varphi \in C_c^\infty(\Omega)$ then the integral over $K = \operatorname{supp}(D^\alpha\varphi)$, for any α, of any continuous or integrable function, is well-defined and finite:

$$g \in L^1(\Omega) \quad \implies \quad \int_\Omega g\, D^\alpha\varphi\, dm = \int_K g\, D^\alpha\varphi\, dm \quad \text{is finite}$$

- we will use $\varphi \in C_c^\infty(\Omega)$ as a *test function* to define a derivative of a much more general kind of function ... if the derivative exists

Definition

if $f \in L^1(\Omega)$, and if $g \in L^1(\Omega)$ satisfies

$$\int_\Omega g\,\varphi\,dm = (-1)^{|\alpha|} \int_\Omega f\,D^\alpha \varphi\,dm \qquad \text{for all } \varphi \in C_c^\infty(\Omega)$$

then we say g is the *αth weak derivative* of f, and we write $D^\alpha f = g$ a.e.

- this definition uses "integration-by-parts 1" for partial derivatives
- because we don't care if functions are (globally) integrable in this definition, we may loosen $f, g \in L^1(\Omega)$ to $f, g \in L^1_{\text{loc}}(\Omega)$

- we will use $\varphi \in C_c^\infty(\Omega)$ as a *test function* to define a derivative of a much more general kind of function ... if the derivative exists

Definition

if $f \in L^1_{\text{loc}}(\Omega)$, and if $g \in L^1_{\text{loc}}(\Omega)$ satisfies

$$\int_\Omega g\,\varphi\,dm = (-1)^{|\alpha|} \int_\Omega f\,D^\alpha\varphi\,dm \qquad \text{for all } \varphi \in C_c^\infty(\Omega)$$

then we say g is the *αth weak derivative* of f, and we write $D^\alpha f = g$ a.e.

- this definition uses "integration-by-parts 1" for partial derivatives
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Definition

if $\alpha \in (\mathbb{Z}^+)^d$ then $\quad D^\alpha = \dfrac{\partial^{|\alpha|}}{\partial {x_1}^{\alpha_1} \dots \partial {x_d}^{\alpha_d}}$

Definition

$$L^1_{\text{loc}}(\Omega) = \left\{ f : \Omega \to \mathbb{R} \,\middle|\, \begin{array}{c} f \text{ is measurable, and} \\ \int_K |f|\, dm < \infty \text{ whenever } K \subset \Omega \text{ is compact} \end{array} \right\}$$

Definition

if $f \in L^1_{\text{loc}}(\Omega)$, and if $g \in L^1_{\text{loc}}(\Omega)$ satisfies

$$\int_\Omega g\, \varphi\, dm = (-1)^{|\alpha|} \int_\Omega f\, D^\alpha \varphi\, dm \qquad \text{for all } \varphi \in C_c^\infty(\Omega)$$

then we say g is the *αth weak derivative* of f, and we write $D^\alpha f = g$ a.e.

weak derivatives: Example 1

- suppose $\Omega \subset \mathbb{R}^d$ is open and $f \in C^k(\Omega)$
- then the usual partial derivative $D^\alpha f$ exists, as long as $|\alpha| \le k$
- integrate the usual derivative $D^\alpha f$ over Λ, using integration-by-parts 1, for each partial derivative in $\alpha = (\alpha_1, \dots, \alpha_d)$, a total of $|\alpha|$ times:

$$\begin{aligned}
\int_\Omega (D^\alpha f)\, \varphi\, dm &= -\int_\Omega D^{\tilde\alpha} f\, D^{(1,0,\dots,0)} \varphi\, dm \\
&= \dots = (-1)^{\alpha_1} \int_\Omega D^{\tilde{\tilde\alpha}} f\, D^{(\alpha_1,0,\dots,0)} \varphi\, dm \\
&= \dots \\
&= (-1)^{|\alpha|} \int_\Omega f\, (D^\alpha \varphi)\, dm
\end{aligned}$$

- thus the weak derivative g exists, namely $g = D^\alpha f$

weak derivatives: Example 2

- let's do a small $\mathbb{R}^1$ example
- I will draw pictures on the right
- consider this continuous function:

$$f(x) = \begin{cases} |x|, & -1 < x < 1 \\ 1, & |x| \ge 1 \end{cases}$$

- one can show that, for the ordinary derivative $D^\alpha = \frac{d}{dx}$, $\alpha = (1)$, we have

$$\int_{\mathbb{R}} g\varphi\, dm = -\int_{\mathbb{R}} f\varphi'\, dm$$

for every $\varphi \in C_c^\infty(\mathbb{R})$, where

$$g(x) = -\mathbb{1}_{(-1,0)}(x) + \mathbb{1}_{(0,1)}(x)$$

- thus: $g = f'$, as a weak derivative, which is no surprise
- note that g is only defined up to a set of measure zero

weak derivatives: Example 3

- part of the concern with weak derivatives is local integrability
- for example, consider this $f \in L^1_{\text{loc}}(\mathbb{R}^2)$:

$$f(x) = \frac{1}{r}$$

on $\mathbb{R}^2$, where $x = (x_1, x_2)$ and $r = (x_1^2 + x_2^2)^{1/2}$
- note f is continuous, except at $x = 0$
- let B_ϵ be the open ball of radius $\epsilon > 0$ centered at the origin
- $f \in L^1_{\text{loc}}(\mathbb{R}^2)$ because this integral gives a finite answer:

$$\int_{B_\epsilon} |f|\, dm = \int_{B_\epsilon} \frac{1}{r}\, dm = \int_0^{2\pi} \int_0^\epsilon \frac{1}{r}\, r\, dr\, d\theta = 2\pi \int_0^\epsilon 1\, dr = 2\pi\epsilon < +\infty$$

- however, the only candidate (a.e.) for the weak derivative is

$$g(x) = (\nabla f)(x) = -\frac{\langle x_1, x_2 \rangle}{r^3},$$

and $g \notin L^1_{\text{loc}}(\mathbb{R}^2)$ because

$$\int_{B_\epsilon} |g|\, dm = \int_{B_\epsilon} \frac{1}{r^2}\, dm = 2\pi \int_0^\epsilon \frac{1}{r}\, dr = +\infty$$

- however, weak derivatives can exist for functions with singularities!
- for example, consider a new $f \in L^1_{\text{loc}}(\mathbb{R}^2)$, which again is continuous, except at $x = 0$:
$$f(x) = \ln(r)$$
- $f \in L^1_{\text{loc}}(\mathbb{R}^2)$ because this integral gives a finite answer:
$$\int_{B_\epsilon} |f|\, dm = \int_{B_\epsilon} |\ln(r)|\, dm = -2\pi \int_0^\epsilon \ln(r)\, r\, dr = \frac{\pi\epsilon^2}{2}(1 - 2\ln(\epsilon)) < +\infty$$
- now the only candidate for the weak derivative is $g(x) = (\nabla f)(x)$, namely
$$g(x) = -\frac{\langle x_1, x_2 \rangle}{r^2},$$
and $g \in L^1_{\text{loc}}(\mathbb{R}^2)$ because
$$\int_{B_\epsilon} |g|\, dm = \int_{B_\epsilon} \frac{1}{r}\, dm = 2\pi \int_0^\epsilon 1\, dr = 2\pi\epsilon < +\infty$$

Outline

lecture content in weeks 10 & 11

- part of this material is only from these slides

to know from slides:

- Lebesgue's FTC on $[a, b]$, with $f \in L^1$ and $F \in AC$
- divergence theorem in $\mathbb{R}^d$
- integration-by-parts in $\mathbb{R}^d$
- multi-index notation for derivatives in $\mathbb{R}^d$, and definition of $C_c^\infty(\Omega)$
- weak derivative in $\mathbb{R}^d$

- . . . but part will be from Chapter 5 in Saxe, sections 5.1 and 5.2

to define:

- linear operator
- bounded linear operator
- operator norm
- dual space   (*I will add this closely-related topic*)

lecture content in weeks 10 & 11

to prove:

- a linear operator is continuous if and only if it is bounded (Theorem 5.2)
- the bounded linear operators $\mathcal{B}(X, Y)$ form a normed vector space under operator norm (Theorem 5.3)
- if Y is complete then $\mathcal{B}(X, Y)$ is complete (Theorem 5.4)
- the dual space is a Banach space
- dual vectors on a Hilbert space can be represented by elements of the Hilbert space (Riesz representation theorem)

- Assignment 6 is posted at `bueler.github.io/fa`